LORA / BLE SENSOR NODE — REV A

hierarchical schematic — revision A

1 Power

Power

File: power.kicad_sch

2 MCU

MCU

File: mcu.kicad_sch

3 Radio

Radio

File: radio.kicad_sch

4 Sensors

Sensors

File: sensors.kicad_sch

6 RF

RF

File: rf.kicad_sch

SHEET MAP

2	Power	solar input, BQ25570 harvester, 3V3 boost, battery NTC
3	MCU	nRF52840, 32 MHz + 32.768 kHz, DCDC1, decoupling, SWD
4	Radio	SX1262, 32 MHz, DC–DC, SPI and control
5	Sensors	BME280 (I2C), switched 3–pin soil probe
6	RF	868 MHz switchless match, harmonic filter, RX balun

Off–grid agricultural sensor node — 868 MHz LoRa + BLE

Solo Engineering

Sheet: /

File: lora–node.kicad_sch

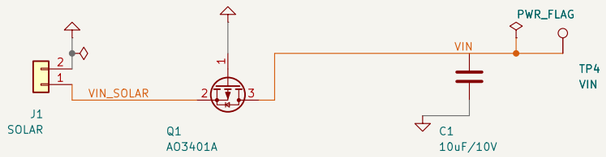
Title: LoRa/BLE Sensor Node — Root

Size: A3	Date: 2026–10–24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/6

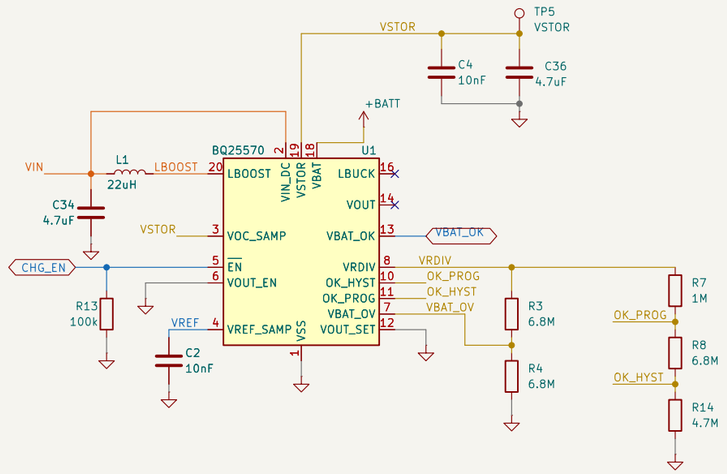
LORA / BLE SENSOR NODE — REV A — POWER

solar harvester — BQ25570 — 3V3 boost — sheet 2 of 5

1 — SOLAR INPUT

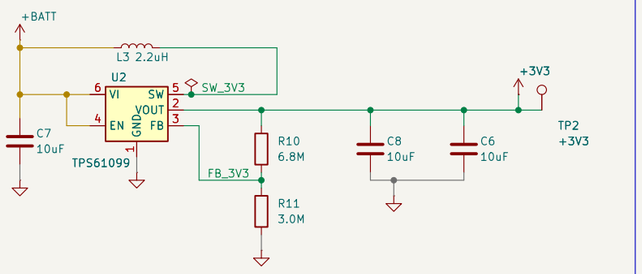


2 — BQ25570 CHARGER & PROGRAMMING DIVIDERS

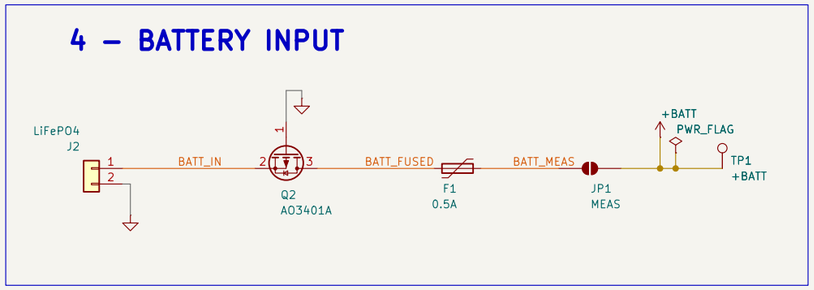


VBAT_OV = 1.5 x 1.21 V x (R3+R4)/R4; VBAT_OK per TI Eq. 3/4. Confirm with SLUC484 before fab.

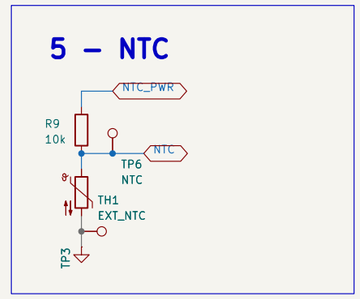
3 — 3V3 BOOST



4 — BATTERY INPUT



5 — NTC



Buck converter unused (spec v1.1): VOUT and LBUCK are no-connect,

VOUT_EN and VOUT_SET are tied to GND.

MPPT: VOC_SAMP tied to VSTOR = 80 % of panel Voc.

Divider strings run VRDIV to GND, total 11–15M (DS 8.3); ratios are first-pass.

NTC_PWR comes from an MCU GPIO so the divider is off by default.

Net classes: orange solar, amber battery, green 3V3, blue sense.

Power tree: solar + BQ25570 harvester + TPS61099 3V3

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Sheet: /Power/
File: power.kicad_sch

Title: LoRa/BLE Sensor Node — Power

Size: A3	Date: 2026-10-24	**Rev: A**
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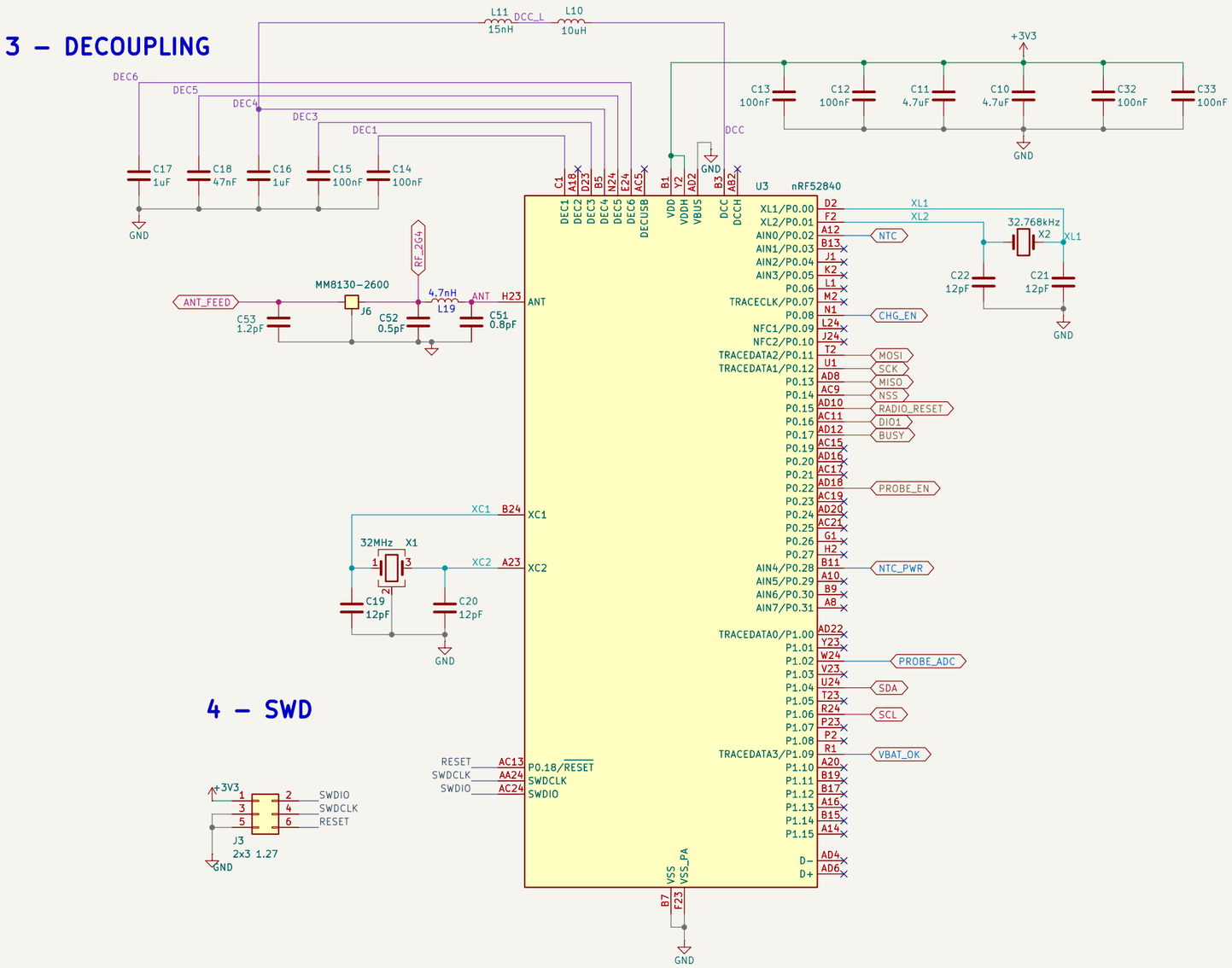
LORA / BLE SENSOR NODE — REV A — MCU

nRF52840 aQFN73 — DCDC1 enabled — normal voltage mode (VDD = VDDH = 3V3)

2 — DCDC1

3 — DECOUPLING

4 — SWD



nRF52840: 32 MHz + 32.768 kHz, DCDC1, decoupling, SWD

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Sheet: /MCU/
File: mcu.kicad_sch

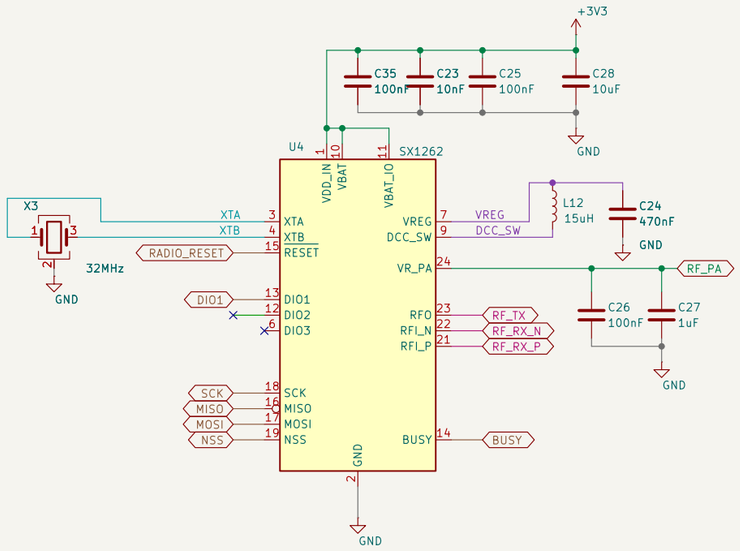
Title: LoRa/BLE Sensor Node — MCU

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/6

LORA / BLE SENSOR NODE — REV A — RADIO

SX1262 — 32 MHz crystal — DCDC — RF front-end on the RF sheet

1 — SX1262



NOTES

SPI / CONTROL

SCK	P0.11	MOSI	P0.12
MISO	P0.13	NSS	P0.14
BUSY	P0.15	DIO1	P0.16
RESET	P0.17		

RF PORTS (SHEET 6)

RF_TX / RF_RX_P / RF_RX_N -> match, filter, balun

ANT 2.4 GHz, from the MCU

CLOCK 32 MHz, XTA / XTB

DC-DC L12 15 uH, C24 470 nF

SX1262 868 MHz: crystal, DC-DC, decoupling, RF ports to sheet 7

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Sheet: /Radio/

File: radio.kicad_sch

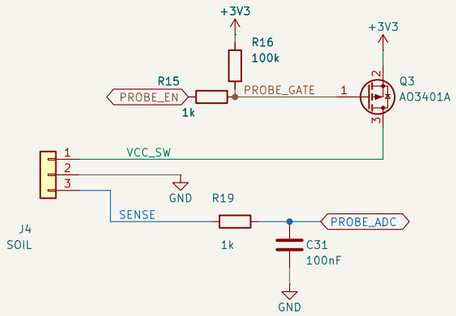
Title: LoRa/BLE Sensor Node — Radio

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/6

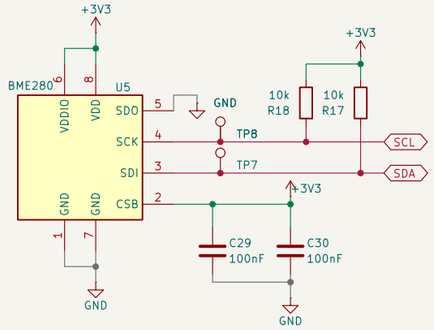
LORA / BLE SENSOR NODE — REV A — SENSORS

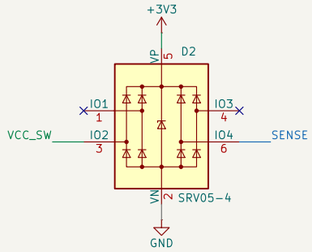
BME280 I2C + switched 3-pin soil probe

2 — SOIL PROBE



1 — BME280





NOTES

U5 = BME280 at 0x76 (SDO low), 100 kHz I2C, 10 k pull-ups
J4 = 1 VCC_SW, 2 GND, 3 SENSE (JST PH, 2.00 mm)
PROBE_EN low turns Q3 on; sample for <= 50 ms at <= 8 mA
R19/C31 = 1 k / 100 nF; D2 clamps VCC_SW and SENSE

BME280 on I2C; soil probe on a switched 3V3 rail with RC + ESD

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Sheet: /Sensors/
File: sensors.kicad_sch

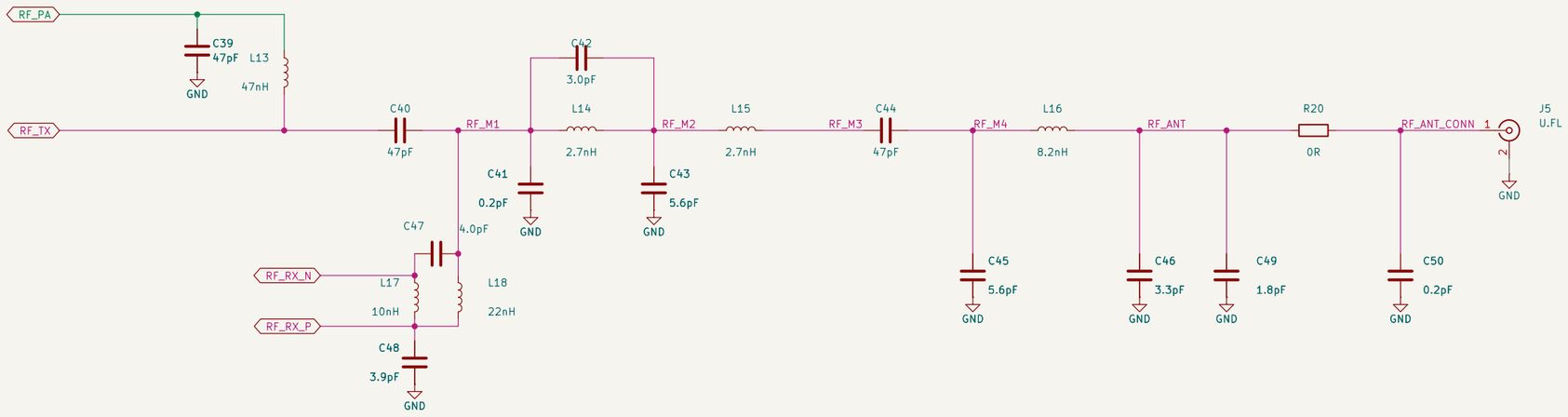
Title: LoRa/BLE Sensor Node — Sensors

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/6

LORA / BLE SENSOR NODE — REV A — RF FRONT-END

868 MHz switchless match, harmonic filter and RX balun — Semtech E460V01A

1 — 868 MHz MATCH & FILTER



NOTES

Semtech E460V01A switchless reference, AN1200.54 Rev 1.1 Table 2.
RFO and the RX balun share the C40/C47 node; no T/R switch.
All 20 positions populated; C41/L18/C49/C50 are simulated values.
R20 is a 0R series link to the 50 ohm U.FL (J5).
VR_PA bulk decoupling C26/C27 stays on the radio sheet.

REFERENCE VALUES

C39 47pF C40 47pF C41 0.2pF C42 3.0pF
C43 5.6pF C44 47pF C45 5.6pF C46 3.3pF
C47 4.0pF C48 3.9pF C49 1.8pF C50 0.2pF
L13 47nH L14 2.7nH L15 2.7nH L16 8.2nH
L17 10nH L18 22nH R20 0R

PORTS

RF_TX RFO, pin 23
RF_PA VR_PA, pin 24
RF_RX_N / RF_RX_P pins 22 / 21
J5 868 MHz U.FL, 50 ohm

TUNING

1. C41/L18/C49/C50 from ngspice: Zopt 16.1+j13.5, 50 ohm.
2. Confirm C49/C50/R20 with a VNA on the first article.
3. Keep the RF trace 50 ohm CPWG, no vias, solid GND.

Switchless 868 MHz front-end for the SX1262 (Semtech E460V01A BOM)
Solo Engineering
Sheet: /RF/
File: rf.kicad_sch
Title: LoRa/BLE Sensor Node — RF Front-End
Size: A3 | Date: 2026-10-24 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 6/6